

CMMI

1. How many maturity levels does CMMI have, and what is the significance of each level?

CMMI has **5 levels**:

- **Level 1 – Initial:** Unpredictable, ad-hoc processes.
- **Level 2 – Managed:** Processes are planned and tracked at the project level.
- **Level 3 – Defined:** Standardized, documented processes at the organizational level.
- **Level 4 – Quantitatively Managed:** Processes measured and controlled using quantitative data.
- **Level 5 – Optimizing:** Continuous improvement based on data and root-cause analysis.

2. What is the difference between a Managed Process (Level 2) and a Defined Process (Level 3)?

- **Level 2:** Each project manages its own processes → not consistent.
- **Level 3:** The organization has **standard processes** applied across all projects
→ consistent and well-documented.

3. Why do organizations aiming for Level 5 need strong investment in process measurement and continuous improvement?

- Because Level 5 requires **data-driven decisions**, identifying root causes, and systematically improving processes → demanding robust measurement and analysis systems.

4. Name one important Process Area from Level 2 and explain why it is necessary.

- **Requirements Management (REQM):**
Ensures requirements are controlled, tracked, and understood, preventing uncontrolled changes and mismatches with customer needs.

5. What is the biggest limitation of CMMI when applied in Agile environments?

- CMMI is **documentation-heavy and process-rigid**, while Agile values flexibility and rapid change → causing conflicts in practice.

ISO (ISO 9001 & ISO 27001)

1. ISO 9001 is based on the PDCA cycle. Which stage do you think has the most impact on output quality?

- **Plan** - because poor planning leads to incorrect requirements, unclear processes, and inconsistent outputs, which directly lowers product/service quality.

2. What is the difference between a Corrective Action and a Preventive Action?

- **Corrective Action:** Fixes the *cause* of a problem **after** it has occurred.
- **Preventive Action:** Eliminates the *cause* of a potential problem **before** it happens.

3. ISO 27001 requires periodic risk assessment. Why must this activity be performed regularly?

- Because threats, vulnerabilities, technologies, and business operations constantly change → regular assessment ensures controls remain effective and risks stay acceptable.

4. Why is the SoA (Statement of Applicability) a mandatory document in ISO 27001?

- Because it lists all security controls, explains which are applied/not applied, and gives justification → provides transparency, accountability, and proof of compliance.

5. List 3 common information security violations that occur in schools/universities.

- Sharing passwords or leaving accounts logged in on public computers.
- Storing student data in unsecured systems or cloud drives.
- Using unauthorized software or downloading infected files, leading to malware attacks.

Software Process (SDLC)

1. What is the difference between Validation and Verification in software development?

- **Verification:** “Are we building the product right?” → checks design, documents, and code.
- **Validation:** “Are we building the right product?” → checks whether the product meets user needs.

2. Why does the Waterfall model struggle when requirements change frequently?

- Because Waterfall is **linear and rigid**; once a phase is completed, changes are costly and require redoing previous phases.

3. In Agile/Iterative models, why is Feedback considered the most important mechanism?

- Feedback ensures **continuous alignment with user needs**, early detection of mistakes, and fast adaptation to change, reducing rework and risk.

4. Which SDLC phase contributes the most to the cost of fixing defects?

- **Maintenance (post - release)** → defects found after deployment are the most expensive due to user impact, debugging complexity, and patching effort.

5. Give an example of a project that failed because it skipped the Requirements Analysis phase.

- **The FBI Virtual Case File (VCF) Project (2000s)** — skipped proper requirements analysis, resulting in unclear needs, massive rework, and eventual cancellation after ~\$170M was spent.

ITIL

1. Distinguish between an Incident and a Problem with a simple example.

- **Incident:** A single unplanned interruption.

Example: A user cannot access email.

- **Problem:** The underlying root cause of one or more incidents.

Example: The email server has a recurring configuration fault causing multiple outages.

2. Why does ITIL focus on delivering service value rather than technology value?

- Because users care about **outcomes**, not tools. Value comes from how services support business goals, not from the technology itself.

3. What is the most important KPI for Incident Management, in your opinion?

- **Mean Time to Restore Service (MTRS)** — it directly reflects how quickly users can return to normal operations.

4. Why does Change Management require a CAB (Change Advisory Board)?

- To evaluate risks, impacts, and business priorities with **multiple expert perspectives**, ensuring changes are safe, controlled, and aligned with organizational needs.

5. What is the role of the Service Desk in improving user experience (UX)?

- It serves as the **single point of contact**, ensuring fast support, clear communication, and a smooth resolution process — all of which significantly improve user satisfaction and trust.

SCRUM

1. What is the main responsibility of the Product Owner in a Scrum project?

- To **maximize product value** by managing and prioritizing the Product Backlog.

2. What is a Sprint Goal, and why does the Scrum team need a clear one?

- A **single, shared objective** for the Sprint.

It keeps the team focused and aligned on what matters most.

3. Why can the Sprint Backlog change during the Sprint but the Sprint Goal cannot?

- Because tasks may be adjusted as the team learns more, but the **objective of the Sprint must remain stable** to keep direction and commitment.

4. How does the Daily Scrum help the team detect problems early?

- By allowing the team to share progress, blockers, and plans daily → issues surface quickly, reducing delays.

5. What is the difference between the Product Backlog and the Sprint Backlog?

- **Product Backlog:** All future product needs, prioritized by the PO.
- **Sprint Backlog:** Items selected for the current Sprint + actionable plan created by the developers

6. Is the Scrum Master the same as a Project Manager? Explain.

No.

- Scrum Master **facilitates**, coaches, and removes impediments; they do **not** assign tasks, manage people, or control scope like a traditional PM.

7. How does the Definition of Done (DoD) improve product quality?

- It sets **clear quality standards** (testing, integration, documentation) → ensures every increment is complete and shippable.

8. Why is the Sprint Review important for stakeholders?

- It provides a chance to **inspect the product**, give feedback, and decide next priorities based on real progress.

9. Scrum allows requirement changes during the project—what are the limits of this flexibility?

Changes are allowed **between Sprints**, but once a Sprint starts:

- The **Sprint Goal** cannot change,
- Large scope changes must wait for the next Sprint.

10. Give an example of a project that may fail without an effective Sprint Retrospective.

- A long-term SaaS product development team that never reflects on workflow issues → technical debt grows, communication breaks down, and velocity drops until the project collapses.